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Data processing device, computer-readable recording medium storing data processing program, and data processing method

Abstract

A data processing device including: a memory storing a plurality of values at a first time point, which include: values of state variables at the first time point and included in an evaluation function of a combinatorial optimization problem with a constraint condition; values of first local fields at the first time point and used to specify a change amount of values of the evaluation function by a change in the values of the state variables; and values of second local fields at the first time point and used to specify an amount to be added to the values of the evaluation function when the constraint condition is not satisfied; and a control circuit configured to return the plurality of state variables, the first local fields, and the second local fields to the values at the first time point when the determination circuit has determined not to accept the change.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based upon and claims the benefit of priority of the prior Japanese Patent Application No. 2023-123028, filed on Jul. 28, 2023, the entire contents of which are incorporated herein by reference.

FIELD

(2) The embodiments discussed herein are related to a data processing device, a non-transitory computer-readable recording medium storing a data processing program, and a data processing method.

BACKGROUND

(3) There is an approach of converting a combinatorial optimization problem into an Ising model that represents a spin behavior of a magnetic body at a time of searching for a solution to the combinatorial optimization problem. Then, a state of the Ising model in which a value of an Ising-type evaluation function, that is, a value corresponding to energy of the Ising model is locally minimized is searched for with the Markov-chain Monte Carlo method. The state in which the minimum value of local minimum values of the evaluation function is reached is treated as an optimal solution. The state variable included in the Ising-type evaluation function is a binary variable that takes a value of 0 or 1. The state variable may be expressed as a bit. Note that a state in which the value of the evaluation function is locally maximized can also be searched for by changing a sign of the evaluation function.

(4) Hereinafter, the Markov-Chain Monte Carlo method will be abbreviated as an MCMC method. In addition, processing based on the MCMC method will also be sometimes referred to as MCMC processing. In the MCMC processing, for example, a state transition is accepted at an acceptance probability of the state transition defined by the Metropolis method or the Gibbs method. Exemplary types of the MCMC method include simulated annealing method and a replica exchange method.

(5) There is a traditional process in which the number of bits for which a change (hereinafter, referred to as a transition) in a state variable is permitted in one trial of a process by the MCMC processing is assumed as one (hereinafter, referred to as a single-bit transition process). In the single-bit transition process, however, when a solution gets stuck at a local solution, it is difficult to escape from the local solution, and the search speed might drop in some cases.

(6) Therefore, there is a suggested process for determining whether or not transitions of a plurality of bits are possible based on a result of computing a change amount of a value of the evaluation function in a case where transitions of the plurality of bits occur and causing transitions of the plurality of bits in a case where the transitions are permitted. Hereinafter, such a process will be referred to as a multiple-bit transition process. By the multiple-bit transition process, escape of a solution from a local solution is facilitated, and the search range may be expanded.

(7) Incidentally, some combinatorial optimization problems have a constraint condition supposed to be satisfied by a solution. For example, a knapsack problem, which is one of the discrete optimization problems, has a constraint condition that a total capacity of luggage that will be packed in a knapsack is equal to or smaller than a capacity of the knapsack.

(8) Japanese Laid-open Patent Publication No. 2020-021209, Japanese Laid-open Patent Publication No. 2020-064536, Japanese Laid-open Patent Publication No. 2021-157361, and Japanese Laid-open Patent Publication No. 2021-165965 are disclosed as related art.

SUMMARY

(9) According to an aspect of the embodiments, there is provided a data processing device including: a memory configured to store a plurality of values at a first time point, the plurality of

values including: values of a plurality of state variables at the first time point, the plurality of state variables being included in an evaluation function of a combinatorial optimization problem with a constraint condition; values of first local fields at the first time point, the first local fields being used to specify a change amount of values of the evaluation function associated with a change in each of the values of the plurality of state variables; and values of second local fields at the first time point, the second local fields being used to specify a constraint violation amount to be added to the values of the evaluation function when the constraint condition is not satisfied; a transition candidate designation circuit configured to calculate the change amount by using the values of the first local fields at the first time point, and designate candidates of which the values are to be changed, one by one from among the plurality of state variables, based on the change amount; an update circuit configured to update the values of the candidates and the values of the first local fields and the second local fields every time the candidates are designated, and further update, at a second time point, the values of the first local fields by using the values at the first time point and the values at the second time point of the second local fields; a cumulative value calculation circuit configured to calculate cumulative values of the change amount when the values of the candidates are changed; a determination circuit configured to determine whether or not to accept the change in the values of a first number of the candidates, which is equal to or greater than two, based on the cumulative values; and a control circuit configured to return each of the values of the plurality of state variables, the values of the first local fields, and the values of the second local fields to the plurality of values at the first time point when the determination circuit has determined not to accept the change in the values of the first number of the candidates.

(10) The object and advantages of the invention will be realized and attained by means of the elements and combinations particularly pointed out in the claims.

(11) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are not restrictive of the invention.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a diagram illustrating an example of a relationship between state variables and second local fields;
- (2) FIG. 2 is a diagram illustrating an example of a change amount of a value of a penalty function due to a change in a value of a state variable;
- (3) FIG. 3 is a diagram illustrating an example of a data processing device according to a first embodiment;
- (4) FIG. 4 is a diagram illustrating an example of updating first local fields and second local fields in a case where a value of $x_{sub,j}$ has changed;
- (5) FIG. 5 is a diagram illustrating a circuit example of a memory, an update circuit, and a transition candidate designation circuit;
- (6) FIG. 6 is a flowchart illustrating a flow of an example of a data processing method according to the first embodiment;
- (7) FIG. 7 is a flowchart illustrating a flow of an example of a single-bit transition process;
- (8) FIG. 8 is a flowchart illustrating a flow of an example of a multiple-bit transition process of the data processing method according to the first embodiment;
- (9) FIG. 9 is a flowchart illustrating a flow of another example of the multiple-bit transition process of the data processing method according to the first embodiment;
- (10) FIG. 10 is a diagram (part 1) illustrating a first example of a pipeline process of the data processing method according to the first embodiment;
- (11) FIG. 11 is a diagram (part 2) illustrating the first example of the pipeline process of the data

processing method according to the first embodiment;

(12) FIG. 12 is a diagram (part 1) illustrating a second example of the pipeline process of the data processing method according to the first embodiment;

(13) FIG. 13 is a diagram (part 2) illustrating the second example of the pipeline process of the data processing method according to the first embodiment;

(14) FIG. 14 is a diagram (part 1) illustrating an example of a data processing device according to a second embodiment;

(15) FIG. 15 is a diagram (part 2) illustrating an example of the data processing device according to the second embodiment;

(16) FIG. 16 is a flowchart illustrating a flow of an example of a multiple-bit transition process of a data processing method according to the second embodiment;

(17) FIG. 17 is a flowchart illustrating a flow of another example of the multiple-bit transition process of the data processing method according to the second embodiment;

(18) FIG. 18 is a diagram (part 1) illustrating an example of a pipeline process of the data processing method according to the second embodiment;

(19) FIG. 19 is a diagram (part 2) illustrating an example of the pipeline process of the data processing method according to the second embodiment;

(20) FIG. 20 is a diagram illustrating a hardware example of an example of the data processing device; and

(21) FIG. 21 is a diagram illustrating a hardware example of a computer.

DESCRIPTION OF EMBODIMENTS

(22) In the traditional multiple-bit transition process, the configuration for computing a change amount of a value of the evaluation function in a case where transitions of a plurality of bits occur, the configuration that propagates data, and the like become complicated. Therefore, implementation has not been easy.

(23) In one aspect, an object of the embodiments is to provide a data processing device, a data processing program, and a data processing method capable of achieving implementation of a multiple-bit transition process of searching for a solution to a combinatorial optimization problem with a constraint condition.

(24) Hereinafter, modes for carrying out the embodiments will be described with reference to the drawings.

(25) A data processing device according to each of the embodiments to be described below searches for a solution to a combinatorial optimization problem with a constraint condition, using an MCMC method such as a simulated annealing method or a replica exchange method, and outputs the solution found by the search.

(26) An evaluation function of the combinatorial optimization problem with a constraint condition is represented by, for example, the following formula.

[MathematicalFormula1]

$$(27) \quad H(x) = -\frac{1}{2} \sum_{i \in D} \sum_{j \in D} W_{ij} x_i x_j - \sum_{i \in D} b_i x_i + \sum_{k \in A} g(h_k) \quad (1)$$

(28) The first and second terms on the right side are the Ising-type evaluation function. The first term on the right side is for integrating products of values (0 or 1) of two state variables and a weighting factor (representing strength of correlation between the two state variables) for all combinations of state variables with neither omission nor duplication. A set of identification numbers of the state variables is represented by D. A state variable with an identification number of i is denoted by $x_{\text{sub}.i}$, and a state variable with an identification number of j is denoted by $x_{\text{sub}.j}$. A weighting factor indicating the magnitude of correlation between the state variables with identification numbers of i and j is denoted by $W_{\text{sub}.ij}$. The second term on the right side is obtained by summing up products of a bias coefficient and a state variable for each identification number. A bias coefficient for the identification number of i is indicated by $b_{\text{sub}.i}$.

(29) The third term on the right side is a constraint term having a value according to the presence or absence of violation of the constraint condition. An identification number of a constraint condition is represented by k , and a set of identification numbers of constraint terms is represented by A . In addition, a proportional coefficient relating to the constraint condition with the identification number= k is denoted by $\gamma_{\text{sub}.k}$. A weight of the constraint term concerning the constraint condition with the identification number= k is represented by $w_{\text{sub}.k}$. Different values may be assigned for $\gamma_{\text{sub}.k}$ for each constraint term. A function according to the type of the constraint condition is denoted by $g(h_{\text{sub}.k})$. Hereinafter, $g(h_{\text{sub}.k})$ will also be sometimes referred to as a penalty function. A local field (LF) used to specify a constraint violation amount added to the value of the evaluation function when the constraint condition with the identification number= k is not satisfied is denoted by $h_{\text{sub}.k}$. Following Formula (2) represents $h_{\text{sub}.k}$. Note that, in the following, $h_{\text{sub}.k}$ will also be sometimes referred to as a second local field in order to be distinguished from a first local field (also sometimes expressed as $h_{\text{sub}.i}$) used to specify the change amount of the value of the evaluation function associated with a change in value of each of a plurality of state variables.

[MathematicalFormula2]

$$(30) \quad h_k = \sum_{i \in D, k \in A} w_{ki} x_i + b_k \quad (2)$$

(31) In Formula (2), a coefficient (weighting factor) indicating the weight of $x_{\text{sub}.i}$ with respect to the constraint condition with the identification number of k is denoted by $w_{\text{sub}.ki}$. A coefficient relating to the constraint condition with the identification number= k is denoted by $b_{\text{sub}.k}$.

(32) FIG. 1 is a diagram illustrating an example of a relationship between state variables and the second local fields.

(33) In FIG. 1, correlations between each of a plurality of state variables ($x_{\text{sub}.1}$, $x_{\text{sub}.i}$, and $x_{\text{sub}.N}$ in the example in FIG. 1), and correlations between each of the plurality of state variables and a plurality of second local fields ($h_{\text{sub}.p}$, $h_{\text{sub}.k}$, and $h_{\text{sub}.r}$ in the example in FIG. 1) are indicated by arrows. The magnitude of these correlations corresponds to $w_{\text{sub}.ij}$ or $w_{\text{sub}.ki}$.

(34) In a case where the constraint condition with the identification number= k is an inequality constraint, $g(h_{\text{sub}.k})$ in Formula (1) can be represented by following Formula (3).

$$(35) \quad g(h_k) = \max[0, h_k] \quad (3)$$

(36) In Formula (3), $\max[0, h_{\text{sub}.k}]$ is a function that outputs a larger value of 0 and $h_{\text{sub}.k}$. In a case where the constraint condition is an inequality constraint, $h_{\text{sub}.k}$ can be represented by following Formula (4).

[MathematicalFormula4]

$$(37) \quad h_k = R_k - U_k = \sum_{i \in D} w_{ki} x_i - U_k \quad (4)$$

(38) A consumption amount (also called a resource amount) of the constraint term with the identification number of k is represented by $R_{\text{sub}.k}$, and an upper limit of the resource amount is represented by $U_{\text{sub}.k}$. The first term on the right side of Formula (2) described above corresponds to $R_{\text{sub}.k}$ in Formula (4), and $+b_{\text{sub}.k}$ in Formula (2) corresponds to $-U_{\text{sub}.k}$ in Formula (4). The difference between $R_{\text{sub}.k}$ and $U_{\text{sub}.k}$ is the constraint violation amount.

(39) Note that another function (such as a step function, for example) can also be used as $g(h_{\text{sub}.k})$.

(40) Hereinafter, the Ising-type evaluation function in the first and second terms on the right side of Formula (1) will also be sometimes expressed as $E(x)$. In addition, the constraint term in the third term on the right side of Formula (1) will also be sometimes expressed as $P(x)$. Using $E(x)$ and $P(x)$, the evaluation function of the combinatorial optimization problem with the constraint condition can be represented as $H(x)=E(x)+P(x)$.

(41) Note that the change amount ($\Delta H_{\text{sub}.i}$) of the value of the evaluation function in a case where

a value of a certain state variable (x.sub.i) changes can be represented by following Formula (5), using the penalty function.

[MathematicalFormula5]

$$(42) \quad H_i = -(\sum_{j \in D} W_{ij} x_j + b_i) x_i + \sum_{k \in A} (g(h_k + W_{ki} x_i) - g(h_k)) \quad (5)$$

(43) In Formula (5), $g(h_{\text{sub.k}} + W_{\text{sub.ki}} \Delta x_{\text{sub.i}}) - g(h_{\text{sub.k}})$ represents a change amount of a value of the penalty function (which can also be referred to as a change amount of the constraint term) in a case where the value of the certain state variable (x.sub.i) changes.

(44) FIG. 2 is a diagram illustrating an example of the change amount of the value of the penalty function due to a change in the value of the state variable. The vertical axis represents the magnitude of $g(h_{\text{sub.k}})$, and the horizontal axis represents $h_{\text{sub.k}}$.

(45) In FIG. 2, $g(h_{\text{sub.k}}) = \max[0, h_{\text{sub.k}}]$ is used as an example of the penalty function. When $h_{\text{sub.k}}$ changes to $h_{\text{sub.k}} + W_{\text{sub.ki}} \Delta x_{\text{sub.i}}$ in association with a change in the value of x.sub.i, the change amount of the value of the penalty function is represented as $g(h_{\text{sub.k}} + W_{\text{sub.ki}} \Delta x_{\text{sub.i}}) - g(h_{\text{sub.k}})$.

(46) As the first local field used to specify the change amount of the value of the evaluation function represented by Formula (5), a variable (h.sub.i) represented by following Formula (6) can be used.

[MathematicalFormula6]

$$(47) \quad h_i = -(\sum_{j \in D} W_{ij} x_j + b_i) - \sum_{k \in A} g(h_k, W_{ki} x_i) \quad (6)$$

(48) In Formula (6), $\Delta g(h_{\text{sub.k}}, W_{\text{sub.ki}} \Delta x_{\text{sub.i}})$ is represented by following Formula (7).

(49) [MathematicalFormula7]

$$g(h_k, W_{ki} x_i) = x_i [g(h_k + W_{ki} x_i) - g(h_k)]$$

(50) As in Formula (7), $\Delta g(h_{\text{sub.k}}, W_{\text{sub.ki}} \Delta x_{\text{sub.i}})$ is an amount that can be calculated from x.sub.i, $h_{\text{sub.k}}$, and $W_{\text{sub.ki}}$.

(51) By using $h_{\text{sub.i}}$ as in Formula (6), $\Delta H_{\text{sub.i}}$ can be represented as $\Delta H_{\text{sub.i}} = -h_{\text{sub.i}} \Delta x_{\text{sub.i}}$.

(52) The data processing device according to each of the embodiments to be described below performs a multiple-bit transition process of searching for a solution to a combinatorial optimization problem with a constraint condition, using the first local field (h.sub.i) as described above. Note that, in the following description, the number of state variables included in the evaluation function as illustrated in Formula (1) is assumed as N, and the number of constraint conditions is assumed as M.

First Embodiment

(53) FIG. 3 is a diagram illustrating an example of a data processing device according to a first embodiment.

(54) A data processing device **10** includes a weighting factor memory **11**, a memory **12**, a transition candidate designation circuit **13**, an update circuit **14**, a cumulative value calculation circuit **15**, a determination circuit **16**, a control circuit **17**, an evaluation function value update-holding circuit **18**, a comparison circuit **19**, and a minimum value update-holding circuit **20**.

(55) The weighting factor memory **11** stores $N \times N$ instances of $W_{\text{sub.ij}}$ and $M \times N$ instances of $W_{\text{sub.ki}}$ included in Formula (6) that is a calculation formula for $h_{\text{sub.i}}$. Note that, since $W_{\text{sub.ki}} = W_{\text{sub.ik}}$ is met, the weighting factor memory **11** does not have to store $N \times M$ instances of $W_{\text{sub.ik}}$. In addition, the weighting factor memory **11** does not have to store $W_{\text{sub.ki}}$ whose value is zero. For example, the weighting factor memory **11** can be formed by a volatile storage device that is an electronic circuit such as a dynamic random access memory (DRAM), a nonvolatile storage device that is an electronic circuit such as a flash memory, or the like. The weighting factor memory **11** may be a storage device such as a hard disk drive (HDD).

(56) Note that the weighting factor memory **11** may store the bias coefficient (b.sub.i), the proportional coefficient ($\gamma_{\text{sub.k}}$), the coefficient (b.sub.k) relating to a constraint condition, data of

other computational conditions, and the like, as well as $W_{sub.ij}$ and $W_{sub.ki}$.

(57) The memory **12** stores the values (hereinafter, also sometimes called states) of the N state variables ($x_{sub.1}$ to $x_{sub.N}$) included in the evaluation function and the values of the first local fields ($h_{sub.1}$ to $h_{sub.N}$) for each of the N state variables. In addition, the memory **12** stores values of the N state variables at a first time point of the solution search and values of the N first local fields at the first time point. Furthermore, the memory **12** stores values of M second local fields ($h_{sub.k}$) and values of the M second local fields at the first time point.

(58) Here, the first time point is, for example, a time point at which the multiple-bit transition process is started (hereinafter, referred to as a start point). In FIG. 3, the values of the N state variables at the start point are expressed as start point states, and the values of the N first local fields at the start point are expressed as start point first local fields. In addition, the values of the M second local fields at the start point are expressed as start point second local fields.

(59) As the memory **12**, a memory having a storage capacity smaller than the storage capacity of the weighting factor memory **11** can be used. For example, the memory **12** can be formed by a volatile storage device that is an electronic circuit such as a static random access memory (SRAM) or a register, or the like. However, the memory **12** is not limited to these storage devices.

(60) Using the values of the N first local fields, the transition candidate designation circuit **13** calculates the change amount of the value of the evaluation function associated with a change in the value of each of the N state variables. The change amount ($\Delta H_{sub.i}$) associated with a change in the value of $x_{sub.i}$ can be represented as $\Delta H_{sub.i} = -h_{sub.i} \Delta x_{sub.i}$. Therefore, the transition candidate designation circuit **13** can calculate $\Delta H_{sub.i}$ by multiplying $h_{sub.i}$ by $-\Delta x_{sub.i}$. The transition candidate designation circuit **13** can compute N instances of $\Delta H_{sub.i}$ in parallel. Furthermore, the transition candidate designation circuit **13** designates, from the start point, a candidate (hereinafter, referred to as a transition candidate) whose value is to be changed, one by one among $x_{sub.1}$ to $x_{sub.N}$, based on the N instances of $\Delta H_{sub.i}$.

(61) In the following example, description will be given on the assumption that the data processing device **10** performs an n (an integer equal to or greater than two)-bit multiple-bit transition process. n can also be called a transition bit depth. The transition candidate designation circuit **13** designates n of the above transition candidates one by one.

(62) The transition candidate designation circuit **13** outputs the identification number of the state variable designated as the transition candidate and the change amount of the value of the evaluation function when the value of the transition candidate is changed. An example of a method for designating a transition candidate and an example of the transition candidate designation circuit **13** will be described later (see FIG. 5).

(63) Every time the transition candidate is designated, the update circuit **14** updates the values of the transition candidates stored in the memory **12** and also updates the values of the N first local fields and the values of the M second local fields. In the following example, it is assumed that the update circuit **14** performs such update until n transition candidates are designated.

(64) For example, in a case where $x_{sub.j}$ is designated as a transition candidate, the update circuit **14** updates $h_{sub.i}$ by adding $\Delta h_{sub.i} = W_{sub.ij} \Delta x_{sub.j}$ to each of the N instances of $h_{sub.i}$. Note that, in a case of $i=j$, $W_{sub.ii}=0$ is met and $h_{sub.i}$ does not change, such that no updating is involved.

(65) In addition, in a case where $x_{sub.j}$ is designated as a transition candidate, the update circuit **14** updates $h_{sub.k}$ by adding $\Delta h_{sub.k} = W_{sub.kj} \Delta x_{sub.j}$ to each of the M instances of $h_{sub.k}$. Note that $h_{sub.k}$ with $W_{sub.kj}=0$ does not change, and thus no updating is involved.

(66) FIG. 4 is a diagram illustrating an example of updating the first local fields and the second local fields in a case where a value of $x_{sub.j}$ has changed. In the example in FIG. 4, the N first local fields are indicated by $h_{sub.1}$ to $h_{sub.N}$, and the M second local fields are indicated by $h_{sub.N+1}$ to $h_{sub.N+M}$. Such computation can be performed by parallel computation with a degree of parallelism of $N+M$.

(67) Furthermore, at a second time point, the update circuit **14** further updates the values of the first local fields, using the values at the first time point and the values at the second time point of the second local fields. The second time point may be when the second local fields are updated. That is, the values of the N first local fields may be updated using the updated second local fields every time the transition candidate is designated. Alternatively, the second time point may be a time point when the transition candidate has been designated a plurality of times. For example, the second time point may be a time point when the transition candidate has been designated n-1 times.

(68) The update of h.sub.i using h.sub.k at the second time point can be represented by, for example, following Formula (8).

[MathematicalFormula8]

$$(69) \quad h_i \leftarrow h_i - \frac{1}{k} [g(h_k, W_{ki}, x_i) - g(h_k^{(old)}, W_{ki}, x_i)] \quad (8)$$

(70) In Formula (8), h.sub.k.sup.(old) represents the second local fields at the first time point stored in the memory **12**. Note that, in a case of i=j, h.sub.i does not change, and thus no updating is involved. Hereinafter, y.sub.k[Δg(h.sub.k, W.sub.kiΔx.sub.i)–Δg(h.sub.k.sup.(old), W.sub.kiΔx.sub.i)] in Formula (8) will also be sometimes expressed as δh.sub.i.

(71) Note that, in the update process as described above, the states (start point states), the values of the first local fields (start point first local fields), and the values of the second local fields (start point second local fields) at the first time point are not updated. The update here is a provisional update, and the updated states and values of the first local fields and the second local fields will be returned to the start point states and the values of the start point first local fields and the start point second local fields in some cases by processing to be described later. The update circuit **14** performs the above update process under the control of the control circuit **17**.

(72) The cumulative value calculation circuit **15** calculates a cumulative value of the change amount of the value of the evaluation function when the value of the transition candidate is changed. The cumulative value calculation circuit **15** can be formed using an adder, a register, or the like, for example.

(73) The determination circuit **16** determines whether or not to accept changes in the values of the n transition candidates, based on the cumulative value of the change amount of the value of the evaluation function, by the MCMC processing. For example, when the cumulative value of the change amount is expressed as ΔE.sub.i, an acceptance probability A.sub.i of accepting changes in the values of the state variables that cause ΔE.sub.i can be represented by Formula (9).

[MathematicalFormula9]

$$(74) \quad A_i = \begin{cases} \min[1, \exp(-\beta E_i)] & \text{MetropolisMethod} \\ 1 / [1 + \exp(\beta E_i)] & \text{GibbsMethod} \end{cases} \quad (9)$$

(75) The reciprocal (β=1/T) of a parameter T (T>0) representing a temperature is denoted by β and is called an inverse temperature. Hereinafter, β and T will also be sometimes referred to as temperature values. The min operator indicates that a minimum value of arguments is taken. The upper right side of Formula (9) corresponds to the Metropolis method. The lower right side of Formula (9) corresponds to the Gibbs method.

(76) The determination circuit **16** makes a determination so as to accept changes in the values of the n transition candidates, for example, at the acceptance probability A.sub.i in Formula (9). The determination circuit **16** that makes such a determination can be formed using a comparator or the like that, for example, outputs a value based on a comparison result between a uniform random number having a value from 0 to 1 and A.sub.i.

(77) In a case where the determination circuit **16** has determined to accept changes in the values of the n transition candidates, the control circuit **17** causes the update circuit **14** to update the values of the N state variables and the values of the first and second local fields stored in the memory **12** to values in a case where the value of the n-th transition candidate has changed. In addition, in a case

where the determination circuit **16** has determined not to accept changes in the values of the n transition candidates, the control circuit **17** returns each of the updated values of the N state variables and values of the first and second local fields to the values at the first time point.

(78) In addition, the control circuit **17** has a function of controlling the operation of each element illustrated in FIG. 1.

(79) For example, the control circuit **17** performs control to cause the update circuit **14** to read $W_{sub.ij}$ or $W_{sub.ki}$ corresponding to identification numbers output by the transition candidate designation circuit **13**, from the weighting factor memory **11**. Furthermore, the control circuit **17** controls the temperature value (T or β) used in the determination process by the determination circuit **16**. In a case where the simulated annealing method is used as the MCMC method, the control circuit **17** gradually decreases the value of T in line with a predetermined schedule, for example. In a case where the replica exchange method is used as the MCMC method, the control circuit **17** performs control such that the processing of each circuit described above is performed independently of each other, for a plurality of replicas of the Ising model. The control circuit **17** sets different temperature values in each of the plurality of replicas and exchanges the temperature values between the replicas at a predetermined exchange probability in a predetermined cycle.

(80) In addition, the control circuit **17** may switch whether to cause the transition candidate designation circuit **13** to execute the single-bit transition process or the multiple-bit transition process, based on the determination result output by the determination circuit **16**. For example, when a determination result indicating that a change in value is not accepted continues for a predetermined period at the time of solution search in the single-bit transition process, the control circuit **17** switches the single-bit transition process to the multiple-bit transition process.

(81) In the single-bit transition process, the transition candidate designation circuit **13** performs a process of determining whether or not to permit a change in value, for each of the N state variables, based on the change amount of the value of the evaluation function. The determination at this time is made in accordance with the MCMC method similarly to the determination circuit **16**. In the multiple-bit transition process, the transition candidate designation circuit **13** performs a process of designating transition candidates one by one from the start point.

(82) When a determination result indicating that a change in value is not accepted continues for a predetermined period at the time of solution search by the multiple-bit transition process, the control circuit **17** may increase the value of n in the multiple-bit transition process. In a case where a determination result indicating that a change in value is accepted is obtained at the time of solution search by the multiple-bit transition process, or in a case where such a determination result continues for a predetermined period, the control circuit **17** may switch the multiple-bit transition process to the single-bit transition process.

(83) In addition, in a case where the determination circuit **16** has determined not to accept changes in the values of the n transition candidates, the control circuit **17** may control the transition candidate designation circuit **13** such that the same n transition candidates are not selected again.

(84) The evaluation function value update-holding circuit **18** includes, for example, a holding circuit such as a register and holds the value of the evaluation function. In a case where the determination circuit **16** has determined to accept changes in the values of the n transition candidates, the evaluation function value update-holding circuit **18** acquires the cumulative value of the change amount of the value of the evaluation function calculated by the cumulative value calculation circuit **15**, updates the value of the evaluation function with the acquired cumulative value, and outputs the updated value.

(85) The comparison circuit **19** compares the value of the evaluation function held in the evaluation function value update-holding circuit **18** with the minimum value of the evaluation function held in the minimum value update-holding circuit **20**.

(86) The minimum value update-holding circuit **20** includes, for example, a holding circuit such as a register and holds the minimum value of the evaluation function obtained until then. In a case

where the comparison circuit **19** outputs a comparison result indicating that the value of the evaluation function held in the evaluation function value update-holding circuit **18** is smaller than the minimum value obtained until then, the minimum value update-holding circuit **20** updates the minimum value with that value of the evaluation function. In addition, the minimum value update-holding circuit **20** holds the values of the N state variables when the minimum value was obtained. The values of the N state variables held when a predetermined end condition is satisfied, for example, are then output as a solution to the combinatorial optimization problem.

(87) FIG. 5 is a diagram illustrating a circuit example of the memory, the update circuit, and the transition candidate designation circuit. Note that FIG. 5 also illustrates an example of storing the weighting factors ($W_{sub.ij}$, $W_{sub.ki}$, and $W_{sub.ik}$) in the weighting factor memory **11**.

(88) The memory **12** includes a state storage area **12a**, a start point state storage area **12b**, a first local field storage area **12c**, a start point first local field storage area **12d**, a second local field storage area **12e**, a start point second local field storage area **12f**, and access control circuits **12g** and **12h**.

(89) The values of the N state variables are stored in the state storage area **12a**, and the values of the N state variables at the first time point (hereinafter, assumed as the start point described above) of the solution search are stored in the start point state storage area **12b**. The values of the N first local fields are stored in the first local field storage area **12c**, and the values of the N first local fields at the start point are stored in the start point first local field storage area **12d**. The values of the M second local fields are stored in the second local field storage area **12e**, and the values of the M second local fields at the start point are stored in the start point second local field storage area **12f**.

(90) The access control circuit **12g** controls reading and writing of the states between the state storage area **12a**, the start point state storage area **12b**, and a state update-holding circuit **14a** of the update circuit **14** under the control of the control circuit **17** illustrated in FIG. 3. Under the control of the control circuit **17**, for example, an operation as described below is performed.

(91) In a case where the state variable is updated, the access control circuit **12g** overwrites the value of the state variable stored in the state storage area **12a** with the updated value of the state variable.

(92) In a case where the determination circuit **16** illustrated in FIG. 3 has determined to accept changes in the values of the n transition candidates, the access control circuit **12g** overwrites the start point states in the start point state storage area **12b** with the values of the N state variables in the state storage area **12a**. Note that the values of the N state variables in the state storage area **12a** are values obtained by updating n values among the values of the N state variables at the start point.

(93) In a case where the determination circuit **16** has determined not to accept changes in the values of the n transition candidates, the access control circuit **12g** overwrites the values of the N state variables in the state storage area **12a** with the values of the N state variables at the start point in the start point state storage area **12b**.

(94) The access control circuit **12h** controls reading and writing of the first local fields between the first local field storage area **12c**, the start point first local field storage area **12d**, and an h.sub.i-holding circuit **14f** of the update circuit **14** under the control of the control circuit **17**. In addition, the access control circuit **12h** controls reading and writing of the second local fields between the second local field storage area **12e**, the start point second local field storage area **12f**, and an h.sub.k-holding circuit **14g** of the update circuit **14** under the control of the control circuit **17**. Under the control of the control circuit **17**, for example, an operation as described below is performed.

(95) In a case where the processing of the transition candidate designation circuit **13** is performed, the access control circuit **12h** writes the values of the N first local fields in the first local field storage area **12c** into the h.sub.i-holding circuit **14f**.

- (96) In a case where the value of the first local field is updated, the access control circuit **12h** overwrites the value of the local field in the first local field storage area **12c** with the updated value of the first local field.
- (97) In a case where the determination circuit **16** has determined to accept changes in the values of the n transition candidates, the access control circuit **12h** overwrites the values of the N first local fields at the start point in the start point first local field storage area **12d** with the values of the N first local fields in the first local field storage area **12c**.
- (98) In a case where the determination circuit **16** has determined not to accept changes in the values of the n transition candidates, the access control circuit **12h** overwrites the values of the N first local fields in the first local field storage area **12c** with the values of the N first local fields at the start point in the start point first local field storage area **12d**.
- (99) In a case where the processing of the transition candidate designation circuit **13** is performed, the access control circuit **12h** writes the values of M auxiliary variables in the second local field storage area **12e** into the h.sub.k-holding circuit **14g**.
- (100) In a case where the value of the second local field is updated, the access control circuit **12h** overwrites the value of the second local field in the second local field storage area **12e** with the updated value of the second local field.
- (101) In a case where the determination circuit **16** has determined to accept changes in the values of the n transition candidates, the access control circuit **12h** overwrites the values of the M second local fields at the start point in the start point second local field storage area **12f** with the values of the M second local fields in the second local field storage area **12e**.
- (102) In a case where the determination circuit **16** has determined not to accept changes in the values of the n transition candidates, the access control circuit **12h** overwrites the values of the M second local fields in the second local field storage area **12e** with the values of the M second local fields at the start point in the start point second local field storage area **12f**.
- (103) At the second time point at which the first local field is updated using the value of the second local field, for example, the access control circuit **12h** writes the values of the M second local fields in the start point second local field storage area **12f** into the h.sub.k-holding circuit **14g**.
- (104) The transition candidate designation circuit **13** includes a ΔH calculation circuit **13a**, a ΔH holding circuit **13b**, a flip-flop (FF) **13c**, an acceptance determination circuit **13d**, and a selection circuit **13e**.
- (105) The ΔH calculation circuit **13a** calculates the change amount of the value of the evaluation function associated with a change in the value of each of the N state variables, using the values of the N first local fields. The change amount ($\Delta H_{\text{sub.i}}$) associated with a change in the value of $x_{\text{sub.i}}$ can be represented as $\Delta H_{\text{sub.i}} = -h_{\text{sub.i}} \Delta x_{\text{sub.i}}$. $\Delta x_{\text{sub.i}}$ can be obtained from the values of $x_{\text{sub.1}}$ to $x_{\text{sub.N}}$ held in the state storage area **12a**.
- (106) Note that the ΔH calculation circuit **13a** may not calculate $\Delta H_{\text{sub.i}}$ for a state variable that has become a transition candidate once while the n transition candidates are designated. This is to avoid a state variable that has once become the transition candidate from becoming the transition candidate again and the value of that state variable from returning to the original state.
- (107) The ΔH holding circuit **13b** holds N instances of $\Delta H_{\text{sub.i}}$ calculated by the ΔH calculation circuit **13a**.
- (108) The FF **13c** holds the value of $h_{\text{sub.k}}$ and delays the timing of outputting $h_{\text{sub.k}}$, thereby supplying $h_{\text{sub.k.sup.(old)}}$ to an h.sub.k-contribution amount calculation circuit **14e** of the update circuit **14**.
- (109) The acceptance determination circuit **13d** accepts a change in the value of at least one state variable. For example, the acceptance determination circuit **13d** outputs the identification number of the state variable that causes $\Delta H_{\text{sub.i}}$ in which the value of the evaluation function becomes the lowest when the value has changed, and that $\Delta H_{\text{sub.i}}$. For example, the acceptance determination circuit **13d** may output the identification numbers of a plurality of state variables that cause

$\Delta H_{\text{sub.i}}$ smaller than a predetermined threshold value, and these instances of $\Delta H_{\text{sub.i}}$. The acceptance determination circuit **13d** may accept changes in the values of one or more state variables in accordance with a predetermined rule.

(110) Note that, in a case where the single-bit transition process is performed, the acceptance determination circuit **13d** accepts a change in the value of each state variable at the acceptance probability indicated by Formula (9).

(111) In a case where the acceptance determination circuit **13d** has accepted changes in the values of a plurality of state variables, the selection circuit **13e** selects one of the plurality of state variables of which the changes in the values have been accepted and outputs the identification number of the selected state variable and $\Delta H_{\text{sub.i}}$ associated with the change in the value of the selected state variable. For example, the selection circuit **13e** may randomly select one state variable or may select one state variable in accordance with a predetermined rule.

(112) The identification number of the state variable selected by the selection circuit **13e** is supplied to the control circuit **17** illustrated in FIG. 3, and $\Delta H_{\text{sub.i}}$ associated with a change in the value of the selected state variable is supplied to the cumulative value calculation circuit **15**.

(113) The update circuit **14** includes the state update-holding circuit **14a**, an $x_{\text{sub.k}}$ -holding circuit **14b**, an $h_{\text{sub.ih.sub.k}}$ update circuit **14c**, an $h_{\text{sub.k}}$ -update detection circuit **14d**, the $h_{\text{sub.k}}$ -contribution amount calculation circuit **14e**, the $h_{\text{sub.i}}$ -holding circuit **14f**, and the $h_{\text{sub.k}}$ -holding circuit **14g**.

(114) The state update-holding circuit **14a** holds the values of the N state variables. In addition, the state update-holding circuit **14a** updates the value of the state variable selected by the selection circuit **13e** under the control of the control circuit **17**.

(115) The $x_{\text{sub.k}}$ -holding circuit **14b** holds M values of $x_{\text{sub.k}}$. For example, $x_{\text{sub.k}}$ can be used to detect whether or not the constraint condition with the identification number= k is satisfied. In a case of the inequality constraint, as illustrated in FIG. 2, $g(h_{\text{sub.k}}) > 0$ is met in a case of $h_{\text{sub.k}} > 0$, and constraint condition violation occurs. Therefore, for example, the $x_{\text{sub.k}}$ -holding circuit **14b** may hold $x_{\text{sub.k}}=1$ when $h_{\text{sub.k}} > 0$ is met and may hold $x_{\text{sub.k}}=0$ when $h_{\text{sub.k}} \leq 0$ is met. However, the data processing device **10** of the present embodiment does not have to use $x_{\text{sub.k}}$. In that case, the $x_{\text{sub.k}}$ -holding circuit **14b** is not involved.

(116) Under the control of the control circuit **17**, the $h_{\text{sub.ih.sub.k}}$ update circuit **14c** updates the values of the first local field ($h_{\text{sub.i}}$) and the second local field ($h_{\text{sub.k}}$) in association with a change in the value of the state variable selected by the selection circuit **13e**. For example, in a case where $x_{\text{sub.j}}$ is a transition candidate, $h_{\text{sub.i}}$ is updated to a value of $h_{\text{sub.i}} = h_{\text{sub.i}} + W_{\text{sub.ij}} \Delta x_{\text{sub.j}}$. In addition, in a case where $x_{\text{sub.j}}$ is designated as a transition candidate, the $h_{\text{sub.ih.sub.k}}$ update circuit **14c** updates $h_{\text{sub.k}}$ to a value of $h_{\text{sub.k}} = h_{\text{sub.k}} + W_{\text{sub.kj}} \Delta x_{\text{sub.j}}$.

(117) The $h_{\text{sub.k}}$ -update detection circuit **14d** detects update of $h_{\text{sub.k}}$. In a case of $W_{\text{sub.kj}}=0$, no update of $h_{\text{sub.k}}$ regarding the constraint condition with the identification number= k is detected.

(118) The $h_{\text{sub.k}}$ -contribution amount calculation circuit **14e** reads non-zero $W_{\text{sub.ki}}$ from the weighting factor memory **11** in order to calculate an amount of change of $h_{\text{sub.i}}$ ($\delta h_{\text{sub.i}}$) due to a change in the value of $h_{\text{sub.k}}$ whose update has been detected. Furthermore, the $h_{\text{sub.k}}$ -contribution amount calculation circuit **14e** acquires $h_{\text{sub.k}}$ from the $h_{\text{sub.k}}$ -holding circuit **14g** and acquires $h_{\text{sub.k.sup.(old)}}$ from the FF **13c**. Then, the $h_{\text{sub.k}}$ -contribution amount calculation circuit **14e** calculates $\delta h_{\text{sub.i}}$ and updates $h_{\text{sub.i}}$ by adding $\delta h_{\text{sub.i}}$ to the original $h_{\text{sub.i}}$.

(119) In a case where the second time point at which the value of $h_{\text{sub.i}}$ is updated based on the value of $h_{\text{sub.k}}$ is a time point when $n-1$ transition candidates have been designated, the value of the second local field at the start point is used as $h_{\text{sub.k.sup.(old)}}$.

(120) The $h_{\text{sub.i}}$ -holding circuit **14f** holds the values of the N first local fields.

(121) The $h_{\text{sub.k}}$ -holding circuit **14g** holds the values of the M second local fields.

(122) Each of the circuits in the data processing device **10** as described above can be formed using an electronic circuit such as an application specific integrated circuit (ASIC) or a field programmable gate array (FPGA), for example. In addition, at least a part of each circuit in the data processing device **10** can also be formed by software processing by a processor executing a program. Examples of the processor include a central processing unit (CPU), a graphics processing unit (GPU), a digital signal processor (DSP), and the like.

(123) Next, a flow of a data processing method (solution search) by the data processing device **10** as described above will be described with reference to a flowchart.

(124) FIG. **6** is a flowchart illustrating a flow of an example of the data processing method according to the first embodiment.

(125) First, initialization is performed under the control of the control circuit **17** (step **S10**). In the processing in step **S10**, initial values of the N state variables, an initial value of the evaluation function, initial values of the N first local fields, and initial values of the M second local fields are set, and an end condition of the solution search, and the like are set. Furthermore, in a case where the simulated annealing method is performed, for example, a start value and an end value of the temperature value, a value by which the temperature value is multiplied for each predetermined temperature shifting cycle, and the like are set. In a case where the replica exchange method is performed, temperature values are set in each replica, a replica exchange cycle and the like are set.

(126) The control circuit **17** determines whether or not to start the multiple-bit transition process (step **S11**). For example, the control circuit **17** causes the single-bit transition process to be started at the time of starting a solution search and causes the multiple-bit transition process to be started in a case where a determination result indicating that a change in value is not accepted continues for a predetermined period.

(127) In a case where the control circuit **17** has determined not to start the multiple-bit transition process, the process in step **S12** is performed. In the process in step **S12**, the single-bit transition process is performed. A flow of the single-bit transition process will be described later (see FIG. 7). After the process in step **S12**, the process in step **S18** is performed.

(128) In a case where the control circuit **17** has determined to start the multiple-bit transition process, the process in step **S13** is performed. In the process in step **S13**, various multiple-bit transition parameters used when the multiple-bit transition process is performed are set. The various multiple-bit transition parameters include n, which is the transition bit depth described above, and the number of repeated trials L, which is the number of times of repeating trials of the multiple-bit transition process. A user may input n and L to the data processing device **10**, or n and L may be preset in a holding circuit such as a register in the data processing device **10**.

Furthermore, the control circuit **17** sets a flag variable Mflip_flag, which has 1 when changes in the values of a plurality of transition candidates is permitted, to 0.

(129) After the process in step **S13**, processes of a multiple-bit transition trial loop (steps **S14** to **S17**) are repeated for $k=1$ to $k<L$.

(130) In the multiple-bit transition trial loop, the multiple-bit transition process is performed (step **S15**). A flow of the multiple-bit transition process will be described later (see FIG. 8).

(131) After the process in step **S15**, the control circuit **17** determines whether or not Mflip_flag has 0 (step **S16**). In a case where the control circuit **17** has determined that Mflip_flag has 0, the control circuit **17** repeats the processes from step **S15** by incrementing k by 1 when $k<L$ is met.

(132) In a case where the control circuit **17** has determined that Mflip_flag does not have 0 or in a case where k reaches L, the control circuit **17** performs the process in step **S18**.

(133) In the process in step **S18**, the control circuit **17** determines whether or not to end the solution search. For example, in a case where a predetermined end condition is satisfied, the control circuit **17** determines to end the solution search, causes a solution to the combinatorial optimization problem, which is a result of the solution search, to be output (step **S19**), and ends the process. In the process in step **S19**, for example, when it is determined to end the solution search, the values of

the N state variables held in the minimum value update-holding circuit **20** are output as a solution to the combinatorial optimization problem. The control circuit **17** may cause a display device (not illustrated) to display the solution or may cause a communication circuit (not illustrated) to transmit the solution to an external device.

(134) In a case where the control circuit **17** has determined not to end the solution search, the control circuit **17** repeats the processes from step **S11**.

(135) FIG. **7** is a flowchart illustrating a flow of an example of the single-bit transition process.

(136) The transition candidate designation circuit **13** calculates $\Delta H_{\text{sub}.i}$ for each of the N state variables, using the current values of the N first local fields (step **S20**).

(137) Then, the transition candidate designation circuit **13** performs acceptance determination (step **S21**). For example, the transition candidate designation circuit **13** accepts a change in the value of each state variable at the acceptance probability indicated by Formula (9), based on $\Delta H_{\text{sub}.i}$ for each of the N state variables and the temperature value supplied from the control circuit **17**.

(138) In a case where changes in the values of a plurality of state variables have been accepted, the transition candidate designation circuit **13** selects one state variable to be updated (step **S22**).

(139) Thereafter, the transition candidate designation circuit **13** determines whether or not the update is allowed (step **S23**). For example, in a case where a change in the value is not accepted for any state variable in the process in step **S21**, it is determined that the update is not allowed.

(140) In a case where the transition candidate designation circuit **13** has determined that the update is allowed, the update circuit **14** performs the update process (step **S24**). In the update process, the value of a state variable allowed to be updated changes. Then, based on that change, the values of the first and second local fields are updated.

(141) At this time, the evaluation function value update-holding circuit **18** updates the held value of the evaluation function, based on $\Delta H_{\text{sub}.i}$ associated with the change in the value of the state variable allowed to be updated. In a case where the updated value of the evaluation function is smaller than the minimum value of the evaluation function held in the minimum value update-holding circuit **20**, the minimum value is updated and held in the minimum value update-holding circuit **20** together with the current values of the N state variables.

(142) In a case where it is determined that the update is not allowed in the process in step **S23**, or after the process in step **S24**, the processing of the data processing device **10** returns to the process in step **S18** illustrated in FIG. **6**.

(143) FIG. **8** is a flowchart illustrating a flow of an example of the multiple-bit transition process of the data processing method according to the first embodiment.

(144) The memory **12** stores the states and the values of $h_{\text{sub}.i}$ and $h_{\text{sub}.k}$ at the start point (step **S30**).

(145) Thereafter, processes of an n-bit transition trial loop to be described below (steps **S31** to **S48**) are repeated for $i=0$ to $i<n$.

(146) In the n-bit transition trial loop, the transition candidate designation circuit **13** selects an object of which $\Delta H_{\text{sub}.i}$ is to be computed (step **S32**). While n transition candidates are designated, a state variable that has become a transition candidate once is excluded from the objects of which $\Delta H_{\text{sub}.i}$ is to be computed. The transition candidate designation circuit **13** calculates $\Delta H_{\text{sub}.i}$ selected as an object to be computed, using $h_{\text{sub}.i}$ stored in the memory **12** (step **S33**).

(147) Then, the transition candidate designation circuit **13** performs acceptance determination (step **S34**). The transition candidate designation circuit **13** accepts a change in the value of at least one state variable.

(148) In a case where changes in the values of a plurality of state variables have been accepted, the transition candidate designation circuit **13** selects one of the plurality of state variables and designates the selected state variable as a transition candidate. In addition, the update circuit **14** updates the state by changing the value of the designated transition candidate (step **S35**). The update of the state here is a provisional update.

(149) After the process in step S35, the cumulative value calculation circuit 15 calculates a cumulative value of $\Delta H_{\text{sub},i}$ when the value of the transition candidate is changed (step S36).

(150) After the process in step S36, the control circuit 17 determines whether or not i has $n-1$ (step S37). That is, it is determined whether or not n transition candidates have been designated. In a case where the control circuit 17 has determined that i has $n-1$, the process in step S38 is performed. In a case where the control circuit 17 has determined that i does not have $n-1$, the process in step S45 is performed.

(151) In the process in step S38, the determination circuit 16 determines whether or not to accept changes in the values of the n transition candidates, based on the cumulative value of $\Delta H_{\text{sub},i}$ in accordance with the MCMC method.

(152) The control circuit 17 determines whether or not the determination circuit 16 has accepted changes in the values of the n transition candidates (step S39). In a case where the control circuit 17 has determined that the determination circuit 16 has accepted changes in the values of the n transition candidates, the control circuit 17 performs the process in step S40. In a case where the control circuit 17 has determined that the determination circuit 16 has not accepted changes in the values of the n transition candidates, the control circuit 17 performs the process in step S43.

(153) In the process in step S40, the control circuit 17 assigns the value of $M_{\text{flip_flag}}$ as 1. Thereafter, the update circuit 14 updates the value of $h_{\text{sub},i}$ (step S41) and updates $h_{\text{sub},k}$ (step S42) in association with a change in the value of the n -th transition candidate.

(154) In the process in step S43, the control circuit 17 assigns the value of $M_{\text{flip_flag}}$ as 0. Thereafter, the control circuit 17 returns the states and the values of $h_{\text{sub},i}$ and $h_{\text{sub},k}$ to the stored states and values at the start point (step S44).

(155) After the process in steps S42 and S44, i is incremented by 1 to be $i=n$, and accordingly, the processing of the data processing device 10 exits the n -bit transition trial loop and returns to the process in step S18 illustrated in FIG. 6.

(156) The update circuit 14 updates the value of $h_{\text{sub},i}$ (step S45) and updates the value of $h_{\text{sub},k}$ (step S46) in association with the change in the value of the transition candidate designated this time. Then, the update circuit 14 further updates the value of $h_{\text{sub},i}$, using $h_{\text{sub},k}$ and $h_{\text{sub},k.\text{sup.}}$ (old) (step S47).

(157) Thereafter, i is incremented by 1, and the n -bit transition trial loop is repeated.

(158) FIG. 9 is a flowchart illustrating a flow of another example of the multiple-bit transition process of the data processing method according to the first embodiment.

(159) The processes in steps S50 to S66 are the same as the processes in steps S30 to S46 illustrated in FIG. 8.

(160) After the process in step S66, the control circuit 17 determines whether or not i has $n-2$ (step S67). That is, it is determined whether or not $n-1$ transition candidates have been designated. In a case where the control circuit 17 has determined that i has $n-2$, the process in step S68 is performed. In a case where the control circuit 17 has determined that i does not have $n-2$, i is incremented by 1, and the n -bit transition trial loop (steps S51 to S69) is repeated.

(161) In the process in step S68, the update circuit 14 updates the value of $h_{\text{sub},i}$, using $h_{\text{sub},k}$ and $h_{\text{sub},k.\text{sup.}}$ (old), which is $h_{\text{sub},k}$ at the start point. Thereafter, i is incremented by 1, and the n -bit transition trial loop (steps S51 to S69) is repeated.

(162) Note that, in the examples in FIGS. 8 and 9, an example in which the acceptance determination is performed based on the cumulative value of $\Delta H_{\text{sub},i}$ for all the n transition candidates has been described, but this example is not restrictive. For example, the acceptance determination may be performed based on the cumulative value of $\Delta H_{\text{sub},i}$ calculated every time a transition candidate is designated.

(163) Furthermore, the order of the processes illustrated in FIGS. 6 to 9 is an example, and the order of the processes may be altered as appropriate.

(164) (Example of Pipeline Process)

(165) The above processes can be performed in parallel for a plurality of replicas, using a pipeline process. The memory **12** stores the values of the N state variables, the values of the N first local fields, and the values of the M second local fields by an amount equal to the number of the plurality of replicas. In addition, the memory **12** may also store the values of the N state variables, the values of the N first local fields, and the values of the M second local fields at the start point by an amount equal to the number of the plurality of replicas, or may store the values common to all the replicas. $W_{sub,ij}$, $W_{sub,ki}$, and $W_{sub,ik}$ stored in the weighting factor memory **11** are common to all the replicas.

(166) The control circuit **17** causes the transition candidate designation circuit **13**, the update circuit **14**, the cumulative value calculation circuit **15**, and the determination circuit **16** to perform the pipeline process for the plurality of replicas. Note that the control circuit **17** may control such that the single-bit transition process is performed for some replicas.

(167) FIGS. **10** and **11** are diagrams illustrating a first example of the pipeline process of the data processing method according to the first embodiment. Note that FIGS. **10** and **11** illustrate an example in which the multiple-bit transition process illustrated in FIG. **8** is performed by the pipeline process.

(168) In FIG. **10**, expressions 0 to R-1 indicate replica numbers of R replicas. In the example in FIG. **10**, trials by multiple-bit transition with a transition bit depth n of three bits are started at the same timing in all the replicas. Note that, since the control circuit **17** can control each replica as an independent process, the start timing of a trial and the value of n may be different among the plurality of replicas. In addition, in the pipeline process, a first-in, first-out (FIFO) is used to adjust operation timings.

(169) Processing performed by each replica is divided into an update phase and a trial phase. The update phase includes read of $W_{sub,ij}$ and $W_{sub,kj}$, calculation of $W_{sub,ij}\Delta x_{sub,j}$ ($=\Delta h_{sub,i}$), $W_{sub,kj}\Delta x_{sub,j}$ ($=\Delta h_{sub,k}$), and $\delta h_{sub,i}$, and the update process for $h_{sub,i}$ and $h_{sub,k}$. The trial phase includes read of $h_{sub,i}$ from the memory **12**, read of $h_{sub,k}$ or $h_{sub,k.sup.(old)}$ from the memory **12**, storage of the states and $h_{sub,i}$ and $h_{sub,k}$ at the start point in the memory **12**, calculation of $\Delta H_{sub,i}$, calculation of the cumulative value of $\Delta H_{sub,i}$, and acceptance determination based on the cumulative value.

(170) The trial phase for designating a first transition candidate starts from a timing t1. Sequentially for each replica, $h_{sub,i}$ is read from the memory **12** and held in the $h_{sub,i}$ -holding circuit **14f**.

(171) The states and $h_{sub,i}$ and $h_{sub,k}$ at the start point are stored in the memory **12** sequentially for each replica from a timing t2. In addition, $\Delta H_{sub,i}$ is calculated sequentially for each replica from the timing t2.

(172) The cumulative value of $\Delta H_{sub,i}$ is calculated sequentially for each replica from a timing t3. Note that, in the example in FIG. **10**, the acceptance determination based on the cumulative value of $\Delta H_{sub,i}$ is not performed until the third transition candidate is designated. Note that, although not illustrated, the acceptance determination based on $\Delta H_{sub,i}$ and the designation of a transition candidate are performed sequentially for each replica from the timing t3. In FIG. **10**, processes in a period (timings t3 to t9) in which the first transition candidate is designated for the R replicas are expressed as a first-bit transition process.

(173) When the first transition candidate is designated sequentially for each replica from the timing t3, the identification number of the transition candidate state variable is supplied to the control circuit **17** via the FIFO. Thereafter, the update phase for the first transition candidate starts (timing t4).

(174) The control circuit **17** causes $W_{sub,ij}$ and $W_{sub,kj}$ corresponding to the identification number of the transition candidate to be read from the weighting factor memory **11** sequentially for each replica from the timing t4. Then, $W_{sub,ij}\Delta x_{sub,j}$ ($=\Delta h_{sub,i}$) and $W_{sub,kj}\Delta x_{sub,j}$ ($=\Delta h_{sub,k}$) are calculated sequentially for each replica from the timing t5. Furthermore, the update process for $h_{sub,i}$ and $h_{sub,k}$ is performed sequentially for each replica from the timing t6.

(175) From the timing t_7 , the trial phase for designating the second transition candidate is started. Sequentially for each replica, $h.sub.i$ is read from the memory **12** and held in the $h.sub.i$ -holding circuit **14f**.

(176) From the timing t_8 , $\Delta H.sub.i$ is calculated sequentially for each replica. In addition, the cumulative value of $\Delta H.sub.i$ is calculated sequentially for each replica from the timing t_9 . In FIG. **10**, processes in a period (timings t_9 to t_{13}) in which the second transition candidate is designated for the R replicas are expressed as a second-bit transition process.

(177) From the timing t_{10} , the update phase associated with the designation of the second transition candidate is started, and processes similar to the processes at the timings t_4 to t_6 are performed.

(178) From the timing t_{11} , the update process for the first local fields using the second local fields is performed. From the timing t_{11} , one of $h.sub.k$ and $h.sub.k.sup.(old)$ is read from the memory **12** and held in the $h.sub.k$ -holding circuit **14g** sequentially for each replica. From the timing t_{12} , the other of $h.sub.k$ and $h.sub.k.sup.(old)$ is read from the memory **12** and held in the $h.sub.k$ -holding circuit **14g** sequentially for each replica. In the example in FIG. **10**, for the replica with the replica number=1, since the values of two second local fields change, the read is performed in two cycles (**a1** and **a2**). In FIG. **10**, processes in a period (timing t_{13} to t_{18}) in which the values of the second local fields are read for the R replicas are expressed as a second local field process.

(179) From the timing t_{14} , the update phase for the first local fields using the second local fields is performed. $W.sub.ki$ corresponding to the identification number= k of $h.sub.k$ changed from the value at the start point is read from the weighting factor memory **11** sequentially for each replica from the timing t_{14} . Then, $\delta h.sub.i$ is calculated sequentially for each replica from the timing t_{15} . Furthermore, the update process for $h.sub.i$ is performed using $\delta h.sub.i$ sequentially for each replica from the timing t_{16} .

(180) From the timing t_{17} , the trial phase for designating the third transition candidate is started, and processes similar to the processes at the timings t_8 and t_9 are performed. In FIG. **11**, processes in a period (timings t_{18} to t_{23}) in which the third transition candidate is designated for the R replicas are expressed as a third-bit transition process.

(181) From the timing t_{19} , the acceptance determination based on the cumulative value of $\Delta H.sub.i$ is performed. In FIG. **11**, the white circle indicates that acceptance is determined to be allowed, and the black rectangle indicates that acceptance is rejected.

(182) In the example in FIG. **11**, replicas with the replica numbers=0 and $R-1$ are determined to be acceptable, and acceptance is rejected for the other replicas. Therefore, the update phase is executed for the replicas with the replica numbers=0 and $R-1$ (timings t_{20} to t_{26}). The update phase is not executed for the other replicas, and the states and $h.sub.i$ and $h.sub.k$ at the start point are read and written into the memory **12** as the current states and $h.sub.i$ and $h.sub.k$.

(183) From the timing t_{21} , the update process for the first local fields using the second local fields is performed. For the replicas with the replica numbers=0 and $R-1$, one of $h.sub.k$ and $h.sub.k.sup.(old)$ is read from the memory **12** and held in the $h.sub.k$ -holding circuit **14g** at the processing timing of each replica. From the timing t_{22} , for the replicas with the replica numbers=0 and $R-1$, the other of $h.sub.k$ and $h.sub.k.sup.(old)$ is read from the memory **12** and held in the $h.sub.k$ -holding circuit **14g** at the processing timing of each replica.

(184) From the timing t_{24} , the update phase for the first local fields using the second local fields is performed. For the replicas with the replica numbers=0 and $R-1$, $W.sub.ki$ corresponding to the identification number= k of $h.sub.k$ changed from the value at the start point is read from the weighting factor memory **11** at the processing timing of each replica from the timing t_{24} . Then, $\delta h.sub.i$ is calculated for these replicas from the timing t_{25} . Furthermore, from the timing t_{26} , the update process for $h.sub.i$ is performed on these replicas, using $\delta h.sub.i$. Note that, for the other replicas, $h.sub.i$ and $h.sub.k$ are updated using $h.sub.i$ and $h.sub.k$ at the start point.

(185) Thereafter, from a timing t_{27} , the trial phase for designating the first transition candidate is

started again, and processes similar to the processes from the timing **t1** are performed.

(186) In a case where the pipeline process as described above is performed, for example, a state indicating a minimum value among the values of the evaluation function obtained while the process for each replica is repeated a predetermined number of trials is output as a solution. Note that the above pipeline process can also be applied to a solution search using the simulated annealing method and can also be applied to a solution search using a replica exchange method.

(187) The control circuit **17** controls a plurality of different replicas so as to perform different processes in the same period by the pipeline process, thereby enabling to process the plurality of replicas simultaneously in parallel.

(188) As illustrated in FIGS. **10** and **11**, one step of each replica process (each process of the trial phase+update phase) is in the same period for all the processes (the first-bit transition process, the second-bit transition process, and the third-bit transition process) that each change the value of one transition candidate. This may enable to conceal a delay in the pipeline of each replica process.

(189) FIGS. **12** and **13** are diagrams illustrating a second example of the pipeline process of the data processing method according to the first embodiment. Note that FIGS. **12** and **13** illustrate an example in which the multiple-bit transition process illustrated in FIG. **9** is performed by the pipeline process.

(190) Processes from a timing **t30** to a timing **t36** are the same as the processes from the timing **t1** to the timing **t7** illustrated in FIG. **10**.

(191) From the timing **t36**, the update process for the first local fields using the second local fields is performed. Processes from the timing **t36** to one cycle before a timing **t43** are the same as the processes from the timing **t11** to the timing **t18** illustrated in FIGS. **10** and **11**.

(192) Processes from one cycle before the timing **t43** to a timing **t47** are the same as the processes from the timing **t19** to the timing **t23** illustrated in FIG. **11**.

(193) Processes from the timing **t47** to a timing **t51** are the same as the processes from the timing **t23** to the timing **t27** illustrated in FIG. **11**.

(194) The processes in FIGS. **12** and **13** are to reflect a difference from the value of the second local field at the start point in the first local field every time the transition candidate is designated.

(195) Note that, after the determination as to whether or not to accept the transition of the n transition candidates, since the above difference of the second local field does not have to be reflected in the first local field except the accepted replicas, the pipeline process period may be shortened.

(196) According to the data processing device **10** and the data processing method of the first embodiment as described above, the states and the values of the first and second local fields at the first time point (the start point in the above example) are stored. Then, the values of the transition candidates designated one by one from that time point are changed to update the first and second local fields, and whether or not to accept changes in the values of these transition candidates is determined based on the cumulative value of $\Delta H_{sub.i}$. Even if the state and the values of the first and second local fields have been updated based on a change in the value of the transition candidate, the state and the values of the first and second local fields return to the state and the values of the first and second local fields at the first time point described above stored in advance, at the time of rejection of the acceptance described above. In this manner, a complicated propagation mechanism or the like for the local fields to achieve multiple-bit transition is no longer involved. This may enable to promote the independence of arithmetic units for each bit, whereby the circuit configuration is simplified, and it becomes easy to implement the multiple-bit transition process. This may also enable large-scale parallelization. Furthermore, even when the number (n) of transition candidates is changed, processing may be performed with the same circuit, and accordingly, the degree of freedom in search may be high.

(197) In addition, since the single-bit transition process and the multiple-bit transition process may be achieved with the same circuit, it may be easy to switch the operation mode.

(198) Furthermore, the pipeline process using a plurality of replicas as illustrated in FIGS. **10** to **13** may also be achieved without complicating the circuit configuration, and high performance may be expected.

(199) In addition, according to the data processing device **10** and the data processing method of the first embodiment, the second local field used to specify the constraint violation amount added to the value of the evaluation function when the constraint condition is not satisfied is used. For the second local field, the value at the first time point is stored in the memory **12** as described above. In addition, the second local field is updated every time the transition candidate is designated. Then, at the second time point, the value of the first local field is further updated using the value of the second local field at the first time point and the value of the second local field at the second time point. In addition, in a case where it is determined not to accept changes in the values of the n transition candidates, the values of the second local fields are returned to the values at the first time point. This may allow the multiple-bit transition process as described above to be applied to a combinatorial optimization problem with a constraint condition.

(200) In addition, for example, as illustrated in FIGS. **9** to **11**, the data processing device **10** may assign the timing of updating the first local fields using the values of the second local fields at the first time point and the second time point, as the time point at which a plurality of transition candidates is designated (in the above example, the time point at which $n-1$ transition candidates are designated). This may allow the changes in the second local fields to be collectively reflected, as compared with a case where the change in the second local field is reflected in the first local field at each time, and accordingly, may reduce the number of arithmetic operations.

(201) Such a data processing device **10** may be expected to be useful as a technique for obtaining an accurate solution at high speed, when solving diverse problems in modern society that can be transformed into a combinatorial optimization problem with a constraint condition.

Second Embodiment

(202) A data processing device according to a second embodiment to be described below divides N state variables into a plurality of state variable groups to execute designation and an update process for n transition candidates, calculation of a cumulative value, and the like in parallel for each state variable group, with a plurality of modules. Note that an example in which N state variables are divided into three state variable groups and processes are performed by three modules will be described below, but these numbers are not restrictive.

(203) FIGS. **14** and **15** are diagrams illustrating an example of the data processing device according to the second embodiment. In FIGS. **14** and **15**, the same elements as the elements illustrated in FIGS. **3** and **5** will be given the same reference signs.

(204) In a data processing device **30** of the second embodiment, a transition candidate designation circuit **31** includes modules **31a1**, **31a2**, and **31a3**. The modules **31a1** to **31a3** designate n transition candidates one by one within ranges of mutually different state variable groups among $x_{\text{sub}.1}$ to $x_{\text{sub}.N}$.

(205) For example, the module **31a1** handles a state variable group of $x_{\text{sub}.1}$ to $x_{\text{sub}.k-1}$, the module **31a2** handles a state variable group of $x_{\text{sub}.k}$ to $x_{\text{sub}.l-1}$, and the module **31a3** handles a state variable group of $x_{\text{sub}.1}$ to $x_{\text{sub}.N}$.

(206) Each of the modules **31a1** to **31a3** is provided with a ΔH calculation circuit **13a**, a ΔH holding circuit **13b**, an acceptance determination circuit **13d**, and a selection circuit **13e** illustrated in FIG. **5**. In FIG. **14**, the ΔH calculation circuit **13a** and the ΔH holding circuit **13b** are collectively expressed as “ ΔH calculation”, the acceptance determination circuit **13d** is expressed as “acceptance”, and the selection circuit **13e** is expressed as “selection”.

(207) The transition candidate designation circuit **31** further includes a selection circuit **31b** and an FF unit **31c**. Under the control of a control circuit **17**, the selection circuit **31b** supplies $\Delta H_{\text{sub}.i}$ due to changes in the values of the transition candidates output from the modules **31a1** to **31a3** to any one of calculation circuits **33a1**, **33a2**, and **33a3** for the cumulative values for the modules

31a1 to 31a3. The identification numbers of the transition candidates are also similarly selected and supplied to any one of holding circuits **33b1**, **33b2**, and **33b3** for the modules **31a1 to 31a3**. The FF unit **31c** includes FFs **13c** illustrated in FIG. 5 as many as the number of modules **31a1 to 31a3**, namely, three FFs. This allows three instances of h.sub.k.sup.(old) to be output in parallel.

(208) The identification numbers and $\Delta H_{\text{sub},i}$ of the first transition candidate output from each of the modules **31a1 to 31a3** may be supplied to one of the calculation circuits **33a1 to 33a3** and one of the holding circuits **33b1 to 33b3** for any module. The identification numbers and $\Delta H_{\text{sub},i}$ of the second and subsequent transition candidates output from each of the modules **31a1 to 31a3** are supplied to the calculation circuits **33a1 to 33a3** and the holding circuits **33b1 to 33b3** for the corresponding modules, respectively.

(209) As the selection circuit **31b** that performs such selection, for example, a 3-to-1 selector with three lanes can be used.

(210) In an update circuit **32**, a state update-holding circuit **32a** holds three state variable groups. The three state variable groups are obtained by dividing the values of the N state variables into three. The state update-holding circuit **32a** can update the values of the state variables selected from each state variable group by the selection circuit **31b** in parallel under the control of the control circuit **17**. In addition, the state update-holding circuit **32a** can output $\Delta x_{\text{sub},j}$ in parallel from each of three holding areas holding the values of the three state variable groups.

(211) In addition, in the update circuit **32**, an h.sub.i.h.sub.k update circuit **32b** updates the values of the first local fields (h.sub.i) and the second local fields (h.sub.k) in association with changes in the values of the three state variables selected by the selection circuit **31b** under the control of the control circuit **17**. For example, in a case where $x_{\text{sub},j}$ is a transition candidate, h.sub.i is updated to a value of $h_{\text{sub},i} = h_{\text{sub},i} + W_{\text{sub},ij} \Delta x_{\text{sub},j}$. In addition, in a case where $x_{\text{sub},j}$ is designated as a transition candidate, h.sub.k is updated to a value of $h_{\text{sub},k} = h_{\text{sub},k} + W_{\text{sub},kj} \Delta x_{\text{sub},j}$.

(212) An h.sub.k-update detection circuit **32d** detects update of three sets of h.sub.k associated with a change in the value of $x_{\text{sub},j}$ in each of the three state variable groups. In a case of $W_{\text{sub},kj} = 0$, no update of h.sub.k regarding the constraint condition with the identification number=k is detected.

(213) An h.sub.k-contribution amount calculation circuit **32c** reads, from a weighting factor memory **11**, non-zero $W_{\text{sub},ki}$ used to calculate an amount of change in h.sub.i ($\delta h_{\text{sub},i}$) due to a change in the value of h.sub.k whose update has been detected, in the three sets of h.sub.k. Furthermore, the h.sub.k-contribution amount calculation circuit **32c** acquires three sets of h.sub.k from an h.sub.k-holding circuit **32f** and acquires h.sub.k.sup.(old) from the FF unit **31c**. Then, the h.sub.k-contribution amount calculation circuit **32c** calculates $\delta h_{\text{sub},i}$ associated with each of changes in the three sets of h.sub.k and updates h.sub.i by adding $\delta h_{\text{sub},i}$ to h.sub.i included in first local field groups corresponding to each set.

(214) An h.sub.i-holding circuit **32e** holds N instances of h.sub.i by dividing the N instances of h.sub.i into three corresponding to three state variable groups. The N/3 instances of h.sub.i included in each of the three first local field groups are updated using the change amount ($\Delta x_{\text{sub},j}$) of $x_{\text{sub},j}$ selected from the corresponding state variable group.

(215) The h.sub.k-holding circuit **32f** holds three sets of M instances of h.sub.k in correspondence with the three modules **31a1 to 31a3**.

(216) A cumulative value calculation circuit **33** in FIG. 15 includes the calculation circuits **33a1 to 33a3** that calculate cumulative values of $\Delta H_{\text{sub},i}$ (expressed as " $\Sigma \Delta H$ ") for each of the modules **31a1 to 31a3**. Furthermore, the cumulative value calculation circuit **33** includes the holding circuits **33b1 to 33b3** that hold the identification numbers of n transition candidates for each of the modules **31a1 to 31a3**, and a selection circuit **33c**.

(217) The selection circuit **33c** selects one of three cumulative values of $\Delta H_{\text{sub},i}$. For example, the selection circuit **33c** selects a cumulative value of $\Delta H_{\text{sub},i}$ that minimizes the value of the evaluation function. The selection circuit **33c** outputs the selected cumulative value and indexes of

the n transition candidates corresponding to the selected cumulative value together.

(218) Based on the selected cumulative value, a determination circuit **16** determines whether or not to accept changes in the values of the n transition candidates corresponding to the selected cumulative value.

(219) Next, a flow of a data processing method (solution search) by the data processing device **30** of the second embodiment will be described with reference to a flowchart.

(220) Since the flows of the entire process and the single-bit transition process are similar to the flows of the processes illustrated in FIGS. **6** and **7**, description will be omitted.

(221) FIG. **16** is a flowchart illustrating a flow of an example of a multiple-bit transition process of the data processing method according to the second embodiment.

(222) Processes in steps **S70** to **S88** in FIG. **16** are substantially similar to the processes in steps **S30** to **S48** illustrated in FIG. **8**.

(223) However, the processes in steps **S72** to **S76** and steps **S85** to **S87** are performed for each module. In addition, in the process in step **S78**, the determination circuit **16** uses one cumulative value selected from among the three cumulative values of $\Delta H_{\text{sub}.i}$ by the selection circuit **33c** to determine whether or not to accept changes in the values of the n transition candidates corresponding to the one cumulative value.

(224) Furthermore, in the process in step **S81**, first, a process of returning the states and the values of the first local fields related to each module to the stored states and values of the first local fields at the start point is performed. This is because the held states and first local fields are different for each module. Thereafter, the control circuit **17** causes the update circuit **32** to update the states and the values of the first local fields related to each module, based on the identification numbers of the n transition candidates held in any one of the holding circuits **33b1** to **33b3**.

(225) In the process in step **S82**, the values of the second local fields held in correspondence with the other modules are also updated so as to match the values of the second auxiliary variables ($h_{\text{sub}.k}$) held in correspondence with the module that has output the selected cumulative value.

(226) In addition, in the process in step **S84**, the control circuit **17** returns the values of the N/3 state variables, the values of the N/3 first local fields, and the values of the M second local fields held in correspondence with each of the three modules **31a1** to **31a3** to the stored values at the start point.

(227) FIG. **17** is a flowchart illustrating a flow of another example of the multiple-bit transition process of the data processing method according to the second embodiment.

(228) Processes in steps **S90** to **S109** in FIG. **17** are substantially similar to the processes in steps **S50** to **S69** illustrated in FIG. **9**.

(229) However, the processes in steps **S92** to **S96** and steps **S105** to **S108** are performed for each module. In addition, in the process in step **S98**, the determination circuit **16** uses one cumulative value selected from among the three cumulative values of $\Delta H_{\text{sub}.i}$ by the selection circuit **33c** to determine whether or not to accept changes in the values of the n transition candidates corresponding to the one cumulative value.

(230) Furthermore, in the process in step **S101**, first, a process of returning the states and the values of the first local fields related to each module to the stored states and values of the first local fields at the start point is performed. Thereafter, the control circuit **17** causes the update circuit **32** to update the states and the values of the first local fields related to each module, based on the identification numbers of the n transition candidates held in any one of the holding circuits **33b1** to **33b3**.

(231) In the process in step **S102**, the values of the second local fields held in correspondence with the other modules are also updated so as to match the values of the second local fields ($h_{\text{sub}.k}$) held in correspondence with the module that has output the selected cumulative value.

(232) In addition, in the process in step **S104**, the control circuit **17** returns the values of the N/3 state variables, the values of the N/3 first local fields, and the values of the M second local fields

held in correspondence with each of the three modules **31a1** to **31a3** to the stored values at the start point.

(233) Note that the order of the processes illustrated in FIGS. **16** and **17** is an example, and the order of the processes may be altered as appropriate.

(234) (Example of Pipeline Process)

(235) Similarly to the data processing of the data processing device **10** of the first embodiment, the above processes can also be performed for a plurality of replicas of the Ising model in parallel, using a pipeline process. The memory **12** stores the values of the N state variables, the values of the N first local fields, and the values of the M second local fields related to each module by an amount equal to the number of the plurality of replicas. In addition, the memory **12** may also store the values of the N state variables, the values of the N first local fields, and the values of the M second local fields at the start point by an amount equal to the number of the plurality of replicas, or may store the values common to all the replicas. In that case, however, in order to hinder the plurality of replicas from simultaneously using the memory **12**, control is performed such that the multiple-bit transition process is not activated in other than one replica in the pipeline process. $W_{sub.ij}$, $W_{sub.ki}$, and $W_{sub.ik}$ stored in the weighting factor memory **11** are common to all the replicas.

(236) The control circuit **17** causes the transition candidate designation circuit **31**, the update circuit **32**, the cumulative value calculation circuit **33**, and the determination circuit **16** to perform the pipeline process for the plurality of replicas. Note that the control circuit **17** may control such that the single-bit transition process is performed for some replicas.

(237) FIGS. **18** and **19** are diagrams illustrating an example of the pipeline process of the data processing method according to the second embodiment. Note that FIGS. **18** and **19** illustrate an example in which the multiple-bit transition process illustrated in FIG. **16** is performed by the pipeline process. The multiple-bit transition process illustrated in FIG. **17** can also be performed by the pipeline process. Hereinafter, differences from the timing chart illustrated in FIG. **12** will be mainly described.

(238) Each process of the update phase is performed for each module. In the trial phase, other than the acceptance determination based on the cumulative value by the determination circuit **16** is performed for each module.

(239) The processes from a timing t_{60} to a timing t_{74} are similar to the processes from the timing t_{30} to the timing t_{44} illustrated in FIG. **12** except that the above-described processes are performed for each module. At timings t_{75} and t_{76} , in the process in step **S81** illustrated in FIG. **16**, a process of returning the states and the values of the first local fields of the replica with the replica number=0 related to each module to the stored states and values of the first local fields at the start point is performed. At timings t_{76} and t_{77} , the update process for the states and the values of the first local fields related to each module in the process in step **S81** illustrated in FIG. **16** and the process corresponding to the process in step **S82** are performed.

(240) By such processes, a standby time occurs between the timing t_{77} and, for example, the start timing (timing t_{78}) of read of $h_{sub.k}$ or $h_{sub.k.sup.(old)}$ to be used to update the first local fields using the second local fields. Note that the control circuit **17** may reduce the standby time by varying the processing cycle, for example.

(241) The second-bit transition process finishes at a timing t_{79} . Thereafter, at timings t_{80} and t_{81} , in the process in step **S81** illustrated in FIG. **16**, a process of returning the states and the values of the first local fields of the replica with the replica number= $R-1$ related to each module to the stored states and values of the first local fields at the start point is performed. At timings t_{81} and t_{82} , the update process for the states and the values of the first local fields related to each module in the process in step **S81** illustrated in FIG. **16** and the process corresponding to the process in step **S82** are performed.

(242) According to the data processing device **30** and the data processing method of the second embodiment as described above, an effect similar to the effect of the data processing device **10** and

the data processing method of the first embodiment may be obtained. Furthermore, since the data processing device **30** of the second embodiment can perform parallel processing by a plurality of modules, a solution search may be performed more efficiently. As described earlier, in the data processing device **10** of the first embodiment, the circuit configuration for achieving the multiple-bit transition process may be simplified. Therefore, even with a configuration in which parallel search using a plurality of modules is performed as in the data processing device **30** of the second embodiment, the circuit configuration may be restrained from becoming complicated.

(243) (Hardware Configuration Example)

(244) FIG. **20** is a diagram illustrating a hardware example of an example of the data processing device.

(245) A data processing device **40** is, for example, a computer and includes a processor **41**, a random access memory (RAM) **42**, an HDD **43**, a GPU **44**, an input interface **45**, a medium reader **46**, a communication interface **47**, and an accelerator card **48**. The units mentioned above are coupled to a bus.

(246) The processor **41** loads at least a part of a program and data stored in the HDD **43** into the RAM **42** and executes the program. Note that the processor **41** may include a plurality of processor cores, for example. In addition, the data processing device **40** may include a plurality of processors. Note that a set of the plurality of processors (multiprocessor) may be called a “processor”. As the processor **41**, a CPU, a GPU, a DSP, or the like can be used.

(247) The RAM **42** is a volatile semiconductor memory that temporarily stores a program to be executed by the processor **41** and data to be used by the processor **41** for arithmetic operations. Note that the data processing device **40** may include a memory of a type other than the RAM **42** or may include a plurality of memories.

(248) The HDD **43** is a nonvolatile storage device that stores software programs such as an operating system (OS), middleware, and application software, and data. The programs include, for example, a program for causing the data processing device **40** to execute a process of searching for a solution in combinatorial optimization. Note that the data processing device **40** may include another type of storage device such as a flash memory or a solid state drive (SSD) or may include a plurality of nonvolatile storage devices.

(249) The GPU **44** outputs an image to a display **44a** coupled to the data processing device **40** in accordance with a command from the processor **41**. As the display **44a**, a cathode ray tube (CRT) display, a liquid crystal display (LCD), a plasma display panel (PDP), an organic electroluminescence (OEL) display, or the like can be used.

(250) The input interface **45** acquires input signals from an input device **45a** coupled to the data processing device **40** and outputs the acquired input signals to the processor **41**. As the input device **45a**, a pointing device such as a mouse, a touch panel, a touch pad, or a trackball, a keyboard, a remote controller, a button switch, or the like can be used. In addition, a plurality of types of input devices may be coupled to the data processing device **40**.

(251) The medium reader **46** is a reading device that reads programs and data recorded on a recording medium **46a**. As the recording medium **46a**, for example, a magnetic disk, an optical disc, a magneto-optical disk (MO), a semiconductor memory, or the like can be used. The magnetic disk includes a flexible disk (FD) and an HDD. The optical disc includes a compact disc (CD) and a digital versatile disc (DVD).

(252) The medium reader **46** copies, for example, a program and data read from the recording medium **46a** to another recording medium such as the RAM **42** or the HDD **43**. The read program is executed by the processor **41**, for example. Note that the recording medium **46a** may be a portable recording medium and is sometimes used for distribution of a program and data. In addition, the recording medium **46a** or the HDD **43** will be sometimes referred to as a computer-readable recording medium.

(253) The communication interface **47** is an interface that is coupled to a network **47a** and

communicates with other information processing devices via the network **47a**. The communication interface **47** may be a wired communication interface coupled to a communication device such as a switch by a cable or a wireless communication interface coupled to a base station by a wireless link.

(254) The accelerator card **48** is a hardware accelerator that searches for a solution to a combinatorial optimization problem. The accelerator card **48** includes an FPGA **48a** and a DRAM **48b**.

(255) Each element illustrated in FIGS. **3**, **14**, and **15** can be formed by the FPGA **48a** and the DRAM **48b**. For example, the weighting factor memory **11** is formed by the DRAM **48b**, and the other circuits and the memory **12** can be formed by the FPGA **48a**.

(256) Note that there may be a plurality of the accelerator cards **48**.

(257) In addition, the processing details described above (such as FIGS. **6** to **9**, **16**, and **17**, for example) performed by the data processing devices **10** and **30** can be achieved by software causing a computer to execute a program.

(258) The program can be recorded in a computer-readable recording medium. As the recording medium, a magnetic disk, an optical disc, a magneto-optical disk, a semiconductor memory, or the like can be used, for example. The magnetic disk includes a flexible disk and an HDD. The optical disc includes a CD, a CD-recordable (R)/rewritable (RW), a DVD, and a DVD-R/RW. The program is sometimes recorded in a portable recording medium and distributed. In that case, the program may be copied from the portable recording medium to another recording medium before execution.

(259) FIG. **21** is a diagram illustrating a hardware example of a computer. The same elements as the elements illustrated in FIG. **20** are given the same reference signs.

(260) A computer **50** is not provided with the accelerator card **48** illustrated in FIG. **20**.

(261) In the computer **50**, a processor **41** executes the above-described processing details (such as FIGS. **6** to **9**, **16**, and **17**, for example) by executing a program. The processor **41** may include a plurality of processor cores in order to execute processes for a plurality of replicas in parallel. In addition, the HDD **43** can be used as the weighting factor memory **11** illustrated in FIG. **3** and the like, and the RAM **42** can be used as the memory **12**.

(262) While one form of the data processing devices, the data processing programs, and the data processing methods according to the embodiments has been described based on the embodiments, this is merely an example and is not limited to the above description.

(263) All examples and conditional language provided herein are intended for the pedagogical purposes of aiding the reader in understanding the invention and the concepts contributed by the inventor to further the art, and are not to be construed as limitations to such specifically recited examples and conditions, nor does the organization of such examples in the specification relate to a showing of the superiority and inferiority of the invention. Although one or more embodiments of the present invention have been described in detail, it should be understood that the various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the invention.

Claims

1. A data processing device comprising: a memory configured to store a plurality of values at a first time point, the plurality of values including: values of a plurality of state variables at the first time point, the plurality of state variables being included in an evaluation function of a combinatorial optimization problem with a constraint condition; values of first local fields at the first time point, the first local fields being used to specify a change amount of values of the evaluation function associated with a change in each of the values of the plurality of state variables; and values of second local fields at the first time point, the second local fields being used to specify a constraint violation amount to be added to the values of the evaluation function when the constraint condition

is not satisfied; a transition candidate designation circuit configured to calculate the change amount by using the values of the first local fields at the first time point, and designate candidates of which the values are to be changed, one by one from among the plurality of state variables, based on the change amount; an update circuit configured to update the values of the candidates and the values of the first local fields and the second local fields every time the candidates are designated, and further update, at a second time point, the values of the first local fields by using the values at the first time point and the values at the second time point of the second local fields; a cumulative value calculation circuit configured to calculate cumulative values of the change amount when the values of the candidates are changed; a determination circuit configured to determine whether or not to accept the change in the values of a first number of the candidates, which is equal to or greater than two, based on the cumulative values; and a control circuit configured to return each of the values of the plurality of state variables, the values of the first local fields, and the values of the second local fields to the plurality of values at the first time point when the determination circuit has determined not to accept the change in the values of the first number of the candidates.

2. The data processing device according to claim 1, wherein the control circuit switches whether to cause the transition candidate designation circuit to perform a first process in which whether or not to permit the change in each of the values of the plurality of state variables is determined based on the change amount, or a second process in which the candidates are designated one by one, and the first time point is a time point when the second process starts.

3. The data processing device according to claim 1, wherein the second time point is a time point when the candidates are designated a plurality of times.

4. The data processing device according to claim 1, wherein the memory stores the values of the plurality of state variables and the values of the first local fields and the second local fields by an amount equal to a number of a plurality of replicas, and the control circuit causes the transition candidate designation circuit, the update circuit, the cumulative value calculation circuit, and the determination circuit to perform a pipeline process for the plurality of replicas.

5. The data processing device according to claim 1, wherein the transition candidate designation circuit includes a plurality of modules, and the plurality of modules designates the first number of the candidates one by one within ranges of mutually different state variable groups among the plurality of state variables, the update circuit updates the values of the state variable groups, the values of the first local fields and the second local fields that correspond to each of the plurality of modules in accordance with the candidates designated by each of the plurality of modules, the cumulative value calculation circuit calculates the cumulative values for each of the plurality of modules, and selects one of the cumulative values calculated for each of the plurality of modules, and the determination circuit determines, based on the selected one of the cumulative values, whether or not to accept the change in the values of the first number of the candidates that correspond to the selected one of the cumulative values.

6. A non-transitory computer-readable recording medium storing a data processing program for causing processor circuitry of a data processing device to execute processing comprising: storing, in a memory of the computer, a plurality of values including: values of a plurality of state variables at a first time point, the plurality of state variables being included in an evaluation function of a combinatorial optimization problem with a constraint condition; values of first local fields at the first time point, the first local fields being used to specify a change amount of values of the evaluation function associated with a change in each of the values of the plurality of state variables; and values of second local fields at the first time point, the second local fields being used to specify a constraint violation amount to be added to the values of the evaluation function when the constraint condition is not satisfied; calculating the change amount by using the values of the first local fields at the first time point, and designating candidates of which the values are to be changed, one by one from among the plurality of state variables, based on the change amount; updating the values of the candidates and the values of the first local fields and the second local fields every

time the candidates are designated, and further updating, at a second time point, the values of the first local fields by using the values at the first time point and the values at the second time point of the second local fields; calculating cumulative values of the change amount when the values of the candidates are changed; determining whether or not to accept the change in the values of a first number of the candidates, which is equal to or greater than two, based on the cumulative values; and returning each of the values of the plurality of state variables, the values of the first local fields, and the values of the second local fields to the plurality of values at the first time point when the determining has determined not to accept the change in the values of the first number of the candidates.

7. A data processing method implemented by processor circuitry of a data processing device, the data processing method comprising: storing, in a memory of the computer, a plurality of values including: values of a plurality of state variables at a first time point, the plurality of state variables being included in an evaluation function of a combinatorial optimization problem with a constraint condition; values of first local fields at the first time point, the first local fields being used to specify a change amount of values of the evaluation function associated with a change in each of the values of the plurality of state variables; and values of second local fields at the first time point, the second local fields being used to specify a constraint violation amount to be added to the values of the evaluation function when the constraint condition is not satisfied; calculating the change amount by using the values of the first local fields at the first time point, and designating candidates of which the values are to be changed, one by one from among the plurality of state variables, based on the change amount; updating the values of the candidates and the values of the first local fields and the second local fields every time the candidates are designated, and further updating, at a second time point, the values of the first local fields by using the values at the first time point and the values at the second time point of the second local fields; calculating cumulative values of the change amount when the values of the candidates are changed; determining whether or not to accept the change in the values of a first number of the candidates, which is equal to or greater than two, based on the cumulative values; and returning each of the values of the plurality of state variables, the values of the first local fields, and the values of the second local fields to the plurality of values at the first time point when the determining has determined not to accept the change in the values of the first number of the candidates.
